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Multimodal explainable AI approach to pediatric appendicitis detection: integrating clinical data and ultrasound images

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Abstract

Background Pediatric appendicitis is a surgical emergency in which early and accurate diagnosis is critical to prevent severe complications. Clinical evaluation is often challenging because symptoms overlap with other abdominal conditions, and routine laboratory tests or imaging findings may be inconclusive. Therefore, improved diagnostic approaches are needed to enhance reliability and consistency.

Methods A multimodal attention fusion framework was developed to integrate patient symptoms, laboratory indicators, and ultrasound images for appendicitis detection. The workflow included data preprocessing, exploratory analysis, model development, training, and validation. Ultrasound images were standardized through resizing and pixel-intensity normalization, before feature extraction using a convolutional neural network, while clinical variables were analyzed as structured inputs. A cross-attention mechanism was employed to fuse multimodal representations and enable final classification.

Results The model was trained and evaluated using the Regensburg Pediatric Appendicitis Dataset. The proposed framework achieved strong diagnostic performance, with 97.3% accuracy, 96.5% precision, 97.8% sensitivity, 96.2% specificity, 97.1% F1-score, and 96.8% AUROC.

Conclusions The multimodal attention-based approach improves diagnostic accuracy and consistency in pediatric appendicitis. By integrating imaging and clinical data and highlighting clinically relevant features, the framework improves predictive performance and aids in interpretability, demonstrating its usefulness as a tool for clinical evaluation.

Keywords Medical decision support, Appendicitis, Multimodal fusion, Cross-attention, ResNet-18, TabNet, Data-driven diagnosis

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Introduction

Appendicitis is a condition in which the appendix, a small pouch that hangs from the beginning of the large intestine, becomes swollen and painful. It is one of the leading reasons for emergency abdominal surgery and is seen most often in children, teenagers, and young adults [1]. The problem begins when the narrow opening of the appendix gets blocked. This happens because of swollen lymphoid tissue, hardened stool (fecalith), intestinal worms, or, less commonly, a tumor. Once the passage is obstructed, bacteria multiply inside, pressure builds, and blood flow is reduced, leading to inflammation that can ultimately cause the appendix to burst [2]. People with appendicitis often complain of pain that starts near the navel and later shifts to the right lower side of the abdomen. Other frequent signs are fever, nausea, vomiting, and an increased white blood cell count. Although appendicitis is very common, diagnosing it quickly is not always simple, as its symptoms may overlap with those of other abdominal illnesses [3]. Delays in identifying appendicitis may cause severe complications. On the other hand, a wrong diagnosis may expose patients to unnecessary surgical procedures. To reduce these risks, clinicians rely on a combination of physical examination, laboratory tests, and imaging, ultrasound being the most common, to support accurate diagnosis and treatment.

The diagnosis of appendicitis has traditionally been carried out using clinical evaluation, laboratory testing, and imaging studies. Clinical decision-making often begins with a physical examination and symptom assessment, supported by scoring systems like Alvarado Score and the Pediatric Appendicitis Score (PAS) [4]. These tools combine factors like abdominal tenderness, white blood cell count, and symptoms like nausea or fever to help estimate the probability of appendicitis [5]. Although clinical scoring systems are useful, they are not always sufficient on their own, particularly when dealing with children or patients who present atypically. In such cases, imaging has become a central part of the diagnostic process [6, 7].

Ultrasound is usually the first test. It is safe, inexpensive, and radiation-free. It is especially helpful for children and pregnant patients [8]. Ultrasound is also widely used because it avoids radiation and is generally safe, which makes it valuable for children [9]. Even so, its accuracy can drop if the operator is less experienced or if factors like body size or bowel gas obscure the view [9, 10]. When more certainty is needed, CT can be used. It provides high sensitivity and specificity. Still, radiation risk limits its use. MRI is another option. It gives detailed images without radiation, but it is more expensive [11, 12].

Alongside imaging, computational tools are being explored. Machine learning (ML) and deep learning (DL) models show promise for appendicitis diagnosis [13, 14]. Simple models such as decision trees and support vector machines (SVMs) use clinical data to estimate risk [15]. Convolutional neural networks (CNNs) work directly with medical images. They can detect signs of appendicitis on CT or ultrasound scans [16]. For instance, Chou et al. [17] trained a CNN to read CT images with accuracy close to radiologists. Kaji et al. [18] showed that deep learning could also assist with ultrasound in children. Despite these advances, imaging alone is not always sufficient, especially with poor-quality scans or unclear clinical signs. In another study, Chowdhury et al. [19] tested a TabNet-based deep learning model for kidney disease in diabetic patients. But its use across different patient groups still needs more validation. Marcinkevičs et al. [20] reported that ML models based on ultrasound can detect appendicitis in children. But these tools often face issues of transparency and practical use in clinics. Imaging findings alone may not fully capture the underlying clinical severity, a multimodal approach is essential.

One useful way to overcome the limitations of single-modality diagnosis is by combining information from different sources. For example, if an ultrasound scan is inconclusive, clinical variables can offer important diagnostic context. Integrating these data streams, known as multimodal fusion, can harness the strengths of both structured records and imaging [21]. Wang et al. [22] has shown that fusing imaging with other biological data, such as genomics, can capture disease patterns that a single data type might miss. The fusion method not only supports more objective decision-making but also reduces the likelihood of diagnostic errors [23, 24]. This approach helps doctors distinguish between simple and severe cases of appendicitis. It also reduces the chance of patients undergoing surgery when it is not needed [22]. When ultrasound findings such as appendix size, wall thickness, and free fluid are combined with lab results, diagnostic accuracy improves [25]. Recently, structured models using hospital records and imaging have been designed to guide surgical planning. Erman et al. [13] found that mixing clinical and imaging data helps predict appendicitis severity in children. However, their work was not tested on outside datasets and showed limited accuracy, which restricts its wider use.

Building on the multimodal fusion approach, we integrate ultrasound images with routine clinical variables to improve preoperative decision-making. The imaging component is analyzed with ResNet-18, while patient data in tabular form is processed using TabNet. Together, these complementary inputs are designed to provide a more robust and reliable diagnostic aid for identifying appendicitis early among diverse patient populations. Unlike previous studies that rely on simple multimodal feature concatenation, the proposed approach employs a cross-attention fusion mechanism to explicitly model interactions between modalities. Cross-attention provides several advantages in this medical setting. First, it captures inter-modality dependencies, enabling the model to emphasize patient-specific clinical features that are most relevant to the visual findings. Second, it enhances interpretability, the learned attention weights can reveal which clinical attributes were most influential for each case [26, 27]. Finally, this strategy has demonstrated superior performance over naive fusion approaches in related multimodal learning literature [28, 29].

Contributions of the study

- Designed a multimodal framework for appendicitis detection that integrates clinical, laboratory, and imaging data.
- Utilized cross-attention to effectively combine heterogeneous inputs and capture complementary information.
- Achieved consistently high diagnostic performance across standard evaluation metrics.
- Identified key predictors through feature importance analysis, with BMI as the strongest contributor, followed by appendicular abscess and thrombocyte count.
- Incorporated explainable AI methods to provide qualitative insights and enhance clinical trust in automated diagnosis.

The remainder of the paper is structured as: "Methods" section describes the proposed methodology, including data preprocessing, exploratory data analysis, and the development of the multimodal predictive framework. The framework integrates ResNet-18 for ultrasound image feature extraction and TabNet for learning from clinical variables, combined through a cross-attention fusion module, along with the training configuration and implementation details. "Results" section presents the experimental results and evaluation. Model performance is assessed using Accuracy, Sensitivity, Specificity, F1-score, AUROC, and calibration measures. This section also includes ablation studies comparing image-only, clinical-only, and late-fusion models, as well as statistical significance testing using McNemar's test. "Discussion" section provides the discussion, interpreting the findings in the context of existing literature and highlighting the clinical implications and relevance of the proposed multimodal framework. It also discusses the limitations of the study. Finally, "Conclusion and future work" section concludes the study by summarizing the main contributions and outlining directions for future research.

Methods

This section describes the complete methodology employed in this study for appendicitis detection. It includes dataset description and data preprocessing, followed by the development of a multimodal deep learning architecture. The Regensburg Pediatric Appendicitis dataset was collected from children admitted with abdominal pain to St. Hedwig Hospital in Regensburg, Germany, between 2016 and 2021 [30]. It contains two types of data: medical images and structured clinical records. It includes:

(i) Ultrasound images: For each paediatric patient, there are between 1 and 15 B-mode grayscale ultrasound scans of the abdominal region. These images typically include views of the lower right quadrant and may show anatomical structures such as the appendix, intestines, lymph nodes, and reproductive organs. Sample ultrasound images are shown in Fig. 1.

(ii) Tabular clinical data: This includes patient demographics (age, gender, body mass index, temperature, white blood cell count etc.), results from laboratory tests, findings from physical examinations, and clinical scoring systems used in appendicitis evaluation, such as the Alvarado and pediatric appendicitis scores. It also features assessments made by radiology experts based on the ultrasound images. Each case in the dataset is annotated with three target labels:

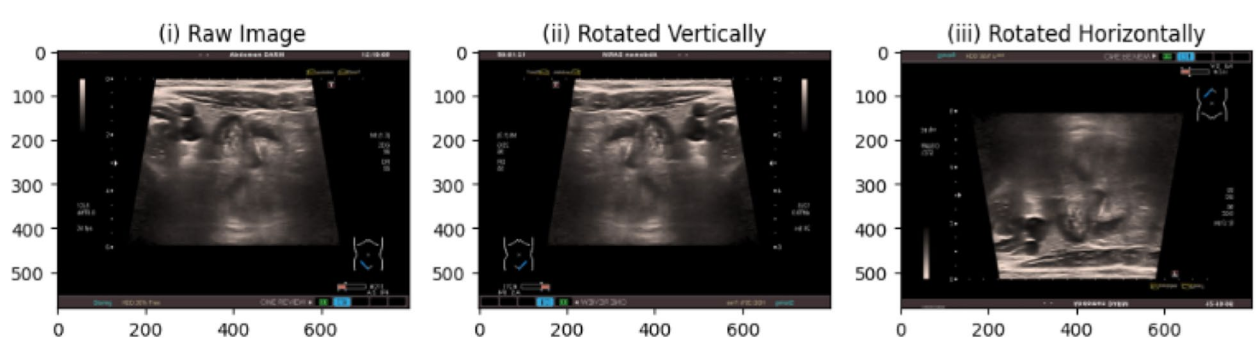


Fig. 1 (i) Raw ultrasound image, (ii) in upright anatomical orientation, and (iii) in sideways anatomical orientation

– Whether the patient was diagnosed with appendicitis or not,
– Whether treatment involved surgery or a conservative approach,
– The severity classification, distinguishing between uncomplicated, complicated, or absence of appendicitis.

The following subsection describes the preprocessing approaches employed in this study.

Preprocessing (for ResNet-18 Input)

Ultrasound images were refined to be used as input to the ResNet-18 model. Each grayscale ultrasound image was expanded to three channels by duplicating single channel data. The frames were then resized to 224×224 pixels, using zero-padding to maintain the original aspect ratio. Intensity values of pixels were normalized by scaling them to the [0,1] range. They were then standardized based on the mean and standard deviation, calculated from the training set only, to avoid information leakage. Only the training images were augmented, using controlled transformations such as small rotations (up to $\pm 10°$), slight shifts (up to 5%), modest zoom changes ($\pm 10\%$), mild brightness adjustments, and low-level Gaussian noise. These operations were chosen to improve model robustness while ensuring that key anatomical structures remained the same. Horizontal flips were avoided to maintain correct orientation. After all image preparation steps, each sample was converted into a $224 \times 224 \times 3$ tensor, making it directly compatible with the ResNet-18 architecture.

80% of the dataset was used for training, while rest was used for testing. To prevent data leakage and overestimation of model performance, each patient in the dataset was assigned a unique patient identifier, and all ultrasound images corresponding to the given patient (ranging from 1 to 15 images per case) were kept entirely within either the training set or the testing set. No images from the same patient appear in both sets. This patient-wise grouping strategy ensures that the model is evaluated on completely unseen patients rather than visually similar images from the same individual. The architecture for prediction of disease is described in the next section.

Multimodal architecture for appendicitis detection

The model consists of two modality-specific encoders followed by a fusion module and a classifier. As seen in Fig. 2, TabNet is used to encode clinical data, with normalization applied to handle both numerical and categorical features effectively.

For imaging data, ResNet-18 serves as the encoder, with region-of-interest (ROI)-based preprocessing applied to highlight the relevant anatomical regions. The encoded vectors are then passed through a cross-attention based fusion module. This module helps the model focus on the most informative features from each source. Finally, a set of dense layers learn the joint representation, and a sigmoid layer outputs the binary classification result, indicating whether appendicitis is present or not. The layered architecture of the proposed model is described with the help of Table 1.

The components detailed in Table 1 are discussed in detail in the following sub-sections.

Image processing component (ResNet-18)

In this framework, ultrasound scans are processed using ResNet-18. ResNet-18 is popular in medical imaging because it offers good accuracy while keeping computation manageable. This makes it suitable for urgent care

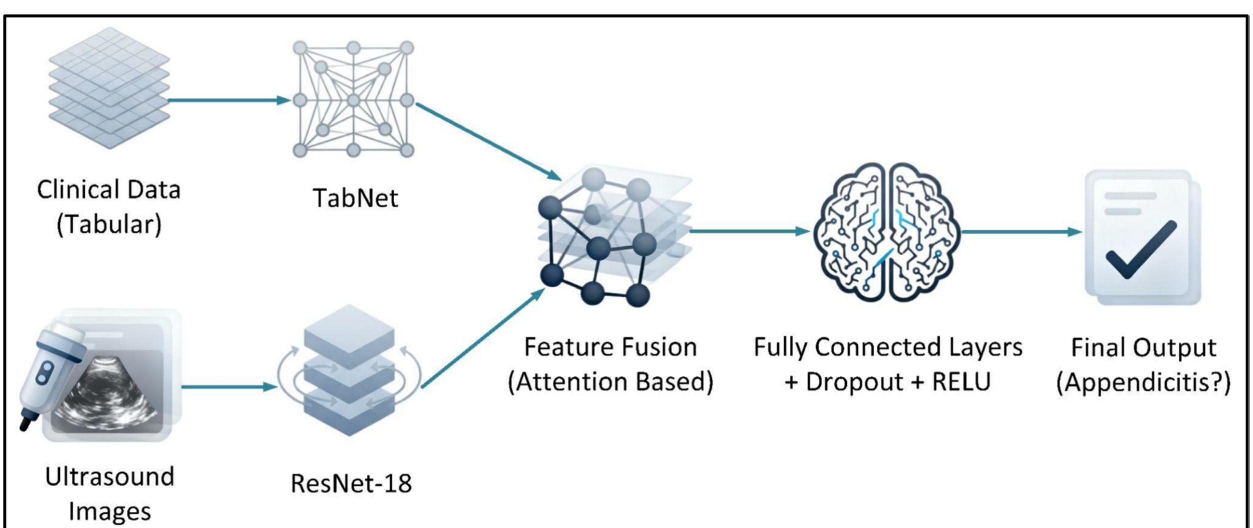


Fig. 2 Model architecture: dual-branch design (Tabnet + ResNet-18) to integrate structured and unstructured data

Table 1 The proposed model consists of components that process image and tabular data, perform feature fusion and predict the disease

Component	Model Type	Purpose
Image Processing	ResNet-18	Extract features from ultrasound images
Tabular Processing	TabNet	Handle clinical/lab data inputs
Feature Fusion	Cross-attention fusion	Image and clinical features
Output	Sigmoid	Predict appendicitis status

Table 2 Architecture of the Resnet18 used for feature extraction from ultrasound images

Layer Type	Output Size	Kernel/Stride	Filters
Conv1	112×112×64	7×7/2	64
MaxPool	56×56×64	3×3/2	-
Residual Block×2	56×56×64	3×3	64
Residual Block×2	28×28×128	3×3/2	128
Residual Block×2	14×14×256	3×3/2	256
Residual Block×2	7×7×512	3×3/2	512
Global Average Pooling Layer	1×1×512	-	-
Fully Connected Layer	512	-	-

Table 3 TabNet architecture for handling the regensburg pediatric appendicitis dataset

Component	Description
Input Features	53 clinical features per patient
Input Dimension	53
Embedding Dimension	64
Decision Steps	5
Feature Transformer	4 layers per step: Fully Connected → BatchNorm → ReLU → Fully Connected
Attentive Transformer	Fully Connected → BatchNorm → Sparsemax activation
Feature Masking	Element-wise multiplication of input features and attention masks
Output Aggregation	Summation of outputs from all decision steps

settings [31]. The network uses shortcut connections, allowing information to skip layers. This helps training run more smoothly and avoids the problem of vanishing gradients. With ultrasound data, ResNet-18 can detect both fine details and broader structures. It captures patterns such as tissue outlines, swelling, and organ shape, all of which may indicate appendicitis. A summary of the model configuration is provided in Table 2.

The network begins with a convolution and max-pooling layer, reducing the input size while increasing depth. It then passes through four stages of residual blocks, each doubling the filter count and halving the spatial dimensions. Finally, global average pooling (GAP) and a fully connected layer output a 512-dimensional feature vector. After processing, the network outputs to the fusion layer, a 512-dimensional feature vector, that summarizes key visual information from the image. This vector is then combined with the non-image feature vector (128-dimensional) from the tabular model in the fusion layer to support a more accurate diagnosis.

TabNet

The tabular data in the Regensburg dataset is handled using TabNet. TabNet is a deep learning architecture specifically designed for tabular data, utilizing sequential attention mechanisms to select and process features dynamically (Table 3).

The model processes 53 clinical features per patient, embedding them into a 64-dimensional space. It uses five decision steps with feature and attentive transformers to learn representations, applying attention-based feature masking. Outputs from all steps are combined and passed through the fusion layer.

Cross-attention-based multimodal fusion

To efficiently combine heterogeneous sources of patient data, ultrasound images and tabular clinical variables, cross-attention based fusion mechanism is adopted. It helps the model focus on important clinical features based on the information extracted from ultrasound images, instead of giving equal importance to all data features. Ultrasound images are first encoded using a pretrained ResNet-18, with the final classification layer removed. This produces the image feature vector:

$$\mathrm{F}_{\mathrm{img}} \in \mathrm{R}^{\mathrm{D}} \quad (1)$$

where D denotes the feature embedding dimension. Simultaneously, clinical variables (age, Alvarado Score, WBC count etc.) are processed using TabNet, which produces a sequence of clinical feature token embeddings:

$$F_{tab} \in R^{N \times D} \quad (2)$$

where N is the number of selected feature tokens, and D is the shared embedding dimension with the image feature. To combine the two modalities, a cross-attention module inspired by Transformer architectures [3] is used. The image feature vector $\mathrm{F}_{\mathrm{img}}$ acts as the query, while the clinical feature embeddings F_{tab} serve as keys and values. The attention mechanism is formulated as:

$$\mathrm{Q} = F_{img} * W_Q \quad (3)$$

$$\mathrm{K} = F_{tab} * W_k \quad (4)$$

$$V = F_{tab} * W_v \quad (5)$$

$$Attention\,(Q, K, V) = softmax\left(\frac{(Q * K^T)}{\sqrt{d_k}}\right) * V \quad (6)$$

Table 4 The output (post-fusion) architecture of the proposed model

Layer	Details
Dense Layer 1	256 neurons + ReLU + Dropout (0.5)
Dense Layer 2	64 neurons + ReLU
Output Layer	Sigmoid

Table 5 Summary of key dimensions of the output layer of the proposed model

Layer	Description	Size
FC1	Dense + ReLU + Dropout (0.3)	64 → 128
FC2	Dense + ReLU + Dropout (0.3)	128 → 64
Output Layer	Sigmoid	64 → 1

Where:

- $W_Q, W_k, W_v \in R^{(D \times d_k)}$ are learnable projection matrices,
- d_k is the attention dimension.

This mechanism computes similarity scores between the image and each clinical feature, allowing the model to generate contextualized, fused representation:

$$F_{fused} \in R^D \tag{7}$$

which captures interactions between imaging and clinical modalities. The fused vector is forwarded to a fully connected classifier, which gives the final prediction: The fusion layer receives the 64-dimensional vector. The post fusion architecture is described using Table 4.

The summary of dimensions of the inputs and the corresponding output is given in Table 5.

This structure allows clinical information to be interpreted alongside visual patterns from the ultrasound image, enabling context-sensitive decision-making. The Results and Discussion sections provide a detailed account of the study's findings, followed by an interpretation of their significance.

Results

This section presents the experimental results obtained from the proposed methodology. First, exploratory data analysis is performed to examine feature distributions and relationships. Following this, the predictive performance of the implemented machine learning models is evaluated and compared. As part of the exploratory data analysis, key demographic and clinical variables including age, gender, severity levels, and body mass index (BMI) were examined to understand the characteristics of the dataset.

Age distribution

The dataset consists of patients who are minors, below 18 years of age, and is slightly left skewed. Maximum cases are observed in mid-childhood and early adolescence, suggesting that the most common age group is 11 to 13 years (Fig. 3).

The patients of the dataset have mean, median and the standard deviation of 11.35 years, 11.44 years and 3.53 years respectively.

BMI distribution

The BMI distribution of the patient dataset is slightly right skewed. The minimum, maximum, mean and median BMI values are 7.83, 38.16, 18.91 and 18.06 respectively (Fig. 4).

Gender distribution

As there are 403 male and 377 female patients, the dataset is fairly balanced (Fig. 5).

The dataset includes 51.7% males and 48.3% females, showing an almost equal gender distribution. This balance reduces bias and supports reliable comparisons between male and female participants.

Severity of cases

The dataset is fairly balanced, with 463 out of 781 patients with appendicitis (51%), while healthy individuals are 317 (49%) (Fig. 6).

As far as severity is concerned, 119 cases are complicated, while 662 individuals are uncomplicated. *Complicated appendicitis* is defined as cases associated with perforation, gangrene, abscess formation or diffuse peritonitis, indicating advanced inflammatory progression, beyond simple (uncomplicated) appendicitis.

Missing data handling

The heatmap of missing data is shown in Fig. 7 (red = present, yellow = missing).

The data values in feature columns with low or moderate missing data are imputed using KNN imputation. The accuracy, precision, recall, specificity, F1 score and AUC of the proposed multimodal architecture is summarized in Table 6.

As seen in Table 6, the model exhibited strong overall performance across all evaluation metrics. Accuracy reached 97.3%, indicating reliable correct classification of most cases. Precision was 96.5%, showing that the majority of predicted positives were truly positive. Sensitivity (recall) was the highest at 97.8%, suggesting excellent ability to detect actual positive cases. Specificity was 96.2%, confirming that the system could also correctly identify negative cases with high confidence. The F1-score of 97.1% reflects a balanced trade-off between precision and recall. Finally, the area under the ROC

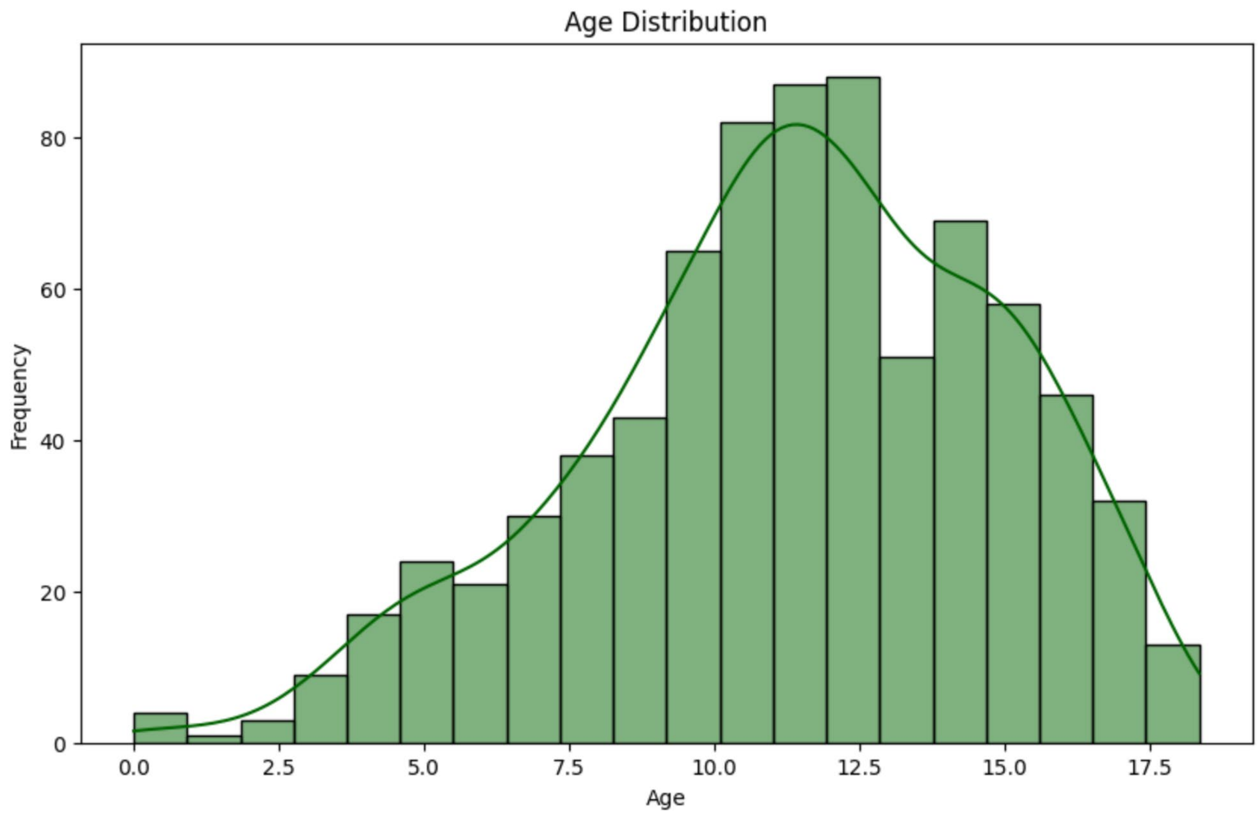


Fig. 3 Age Distribution of the patients whose data was used in this study

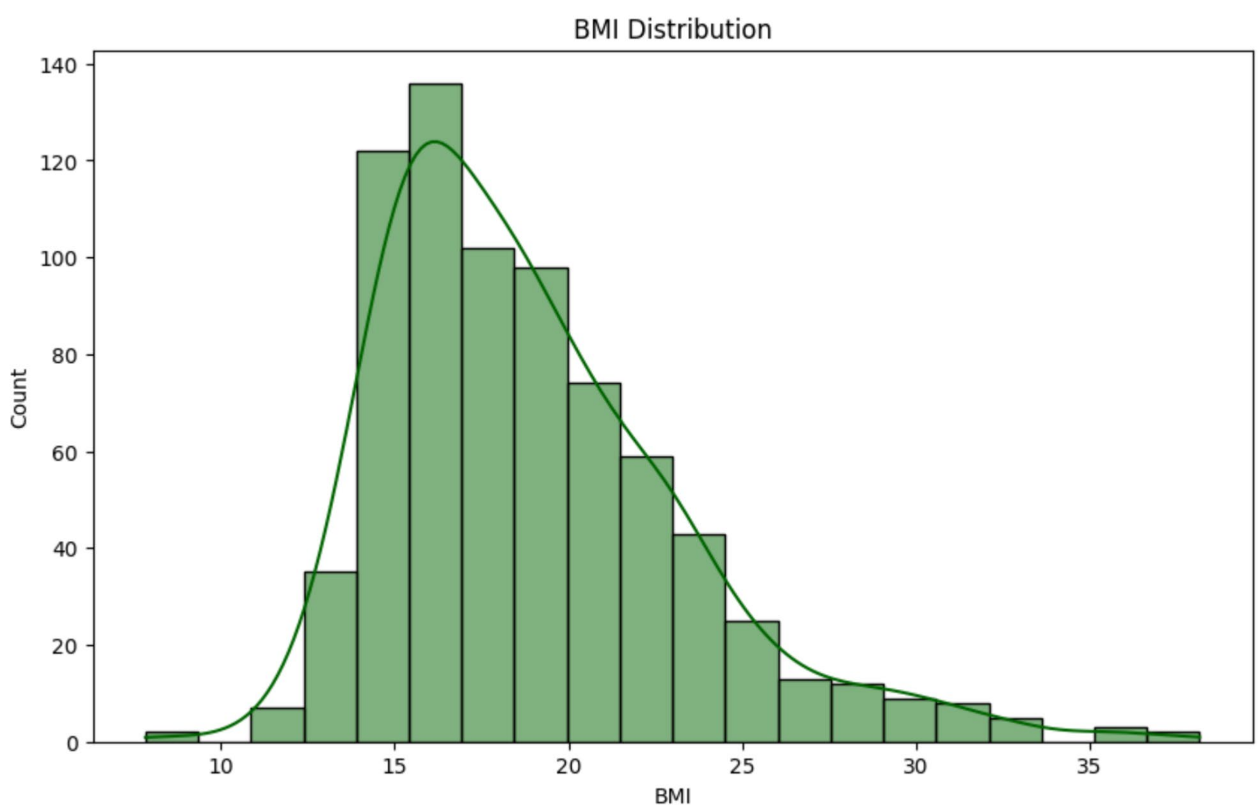


Fig. 4 Body Mass Index (BMI) Distribution of the patients whose data was used in the study

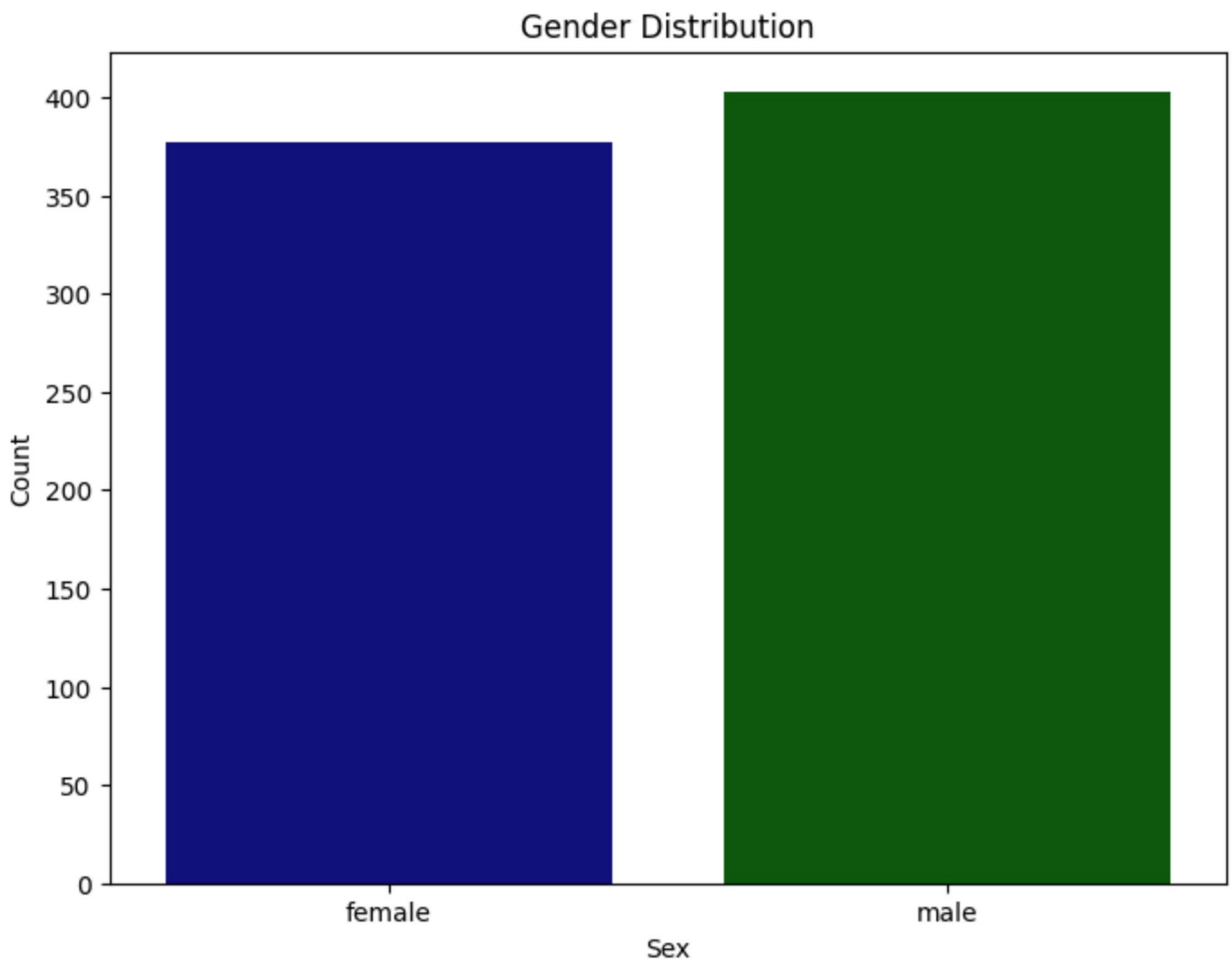


Fig. 5 Gender distribution of the patients whose data was used in the study

curve (AUC) was 96.8%, demonstrating that the model maintains a strong discriminative ability between classes. Collectively, the model demonstrated strong generalization performance, correctly identifying both complicated and uncomplicated cases of appendicitis across a balanced test set. To find the contribution of each component of the architecture, an ablation study was performed, as discussed in the next sub-section.

Contribution of each component

An ablation study was conducted to infer the impact of each model component. Removing either the image or clinical input, or replacing the cross-attention fusion with simple concatenation, resulted in reduced performance. The contribution of each module to the overall accuracy and robustness of the architecture is summarized in Table 7.

The results indicate that both modalities contribute complementary information, and the cross-attention fusion improves their synergy.

Model interpretability

It is the ability to understand and explain how a model arrives at its predictions. Methods such as feature importance for tabular data and Grad-CAM for images help reveal which inputs or regions most influence the output, providing transparency and supporting trust in the model's decisions. The tools applied to aid in clinical transparency are discussed in the following sub-sections.

Clinical feature importance

To understand model behavior and provide insight into the most critical predictors, 53 clinical features were ranked using TabNet's built-in mask-based importance (Fig. 8).

As seen in Fig. 8, the features that have the most significant contribution are WBC count, pediatric appendicitis score, and body temperature. These features directly reflect inflammatory and clinical severity markers, consistent with established pediatric diagnostic practice. Additional features, including age, length of stay, Alvarado score, and ultrasound-based findings, showed

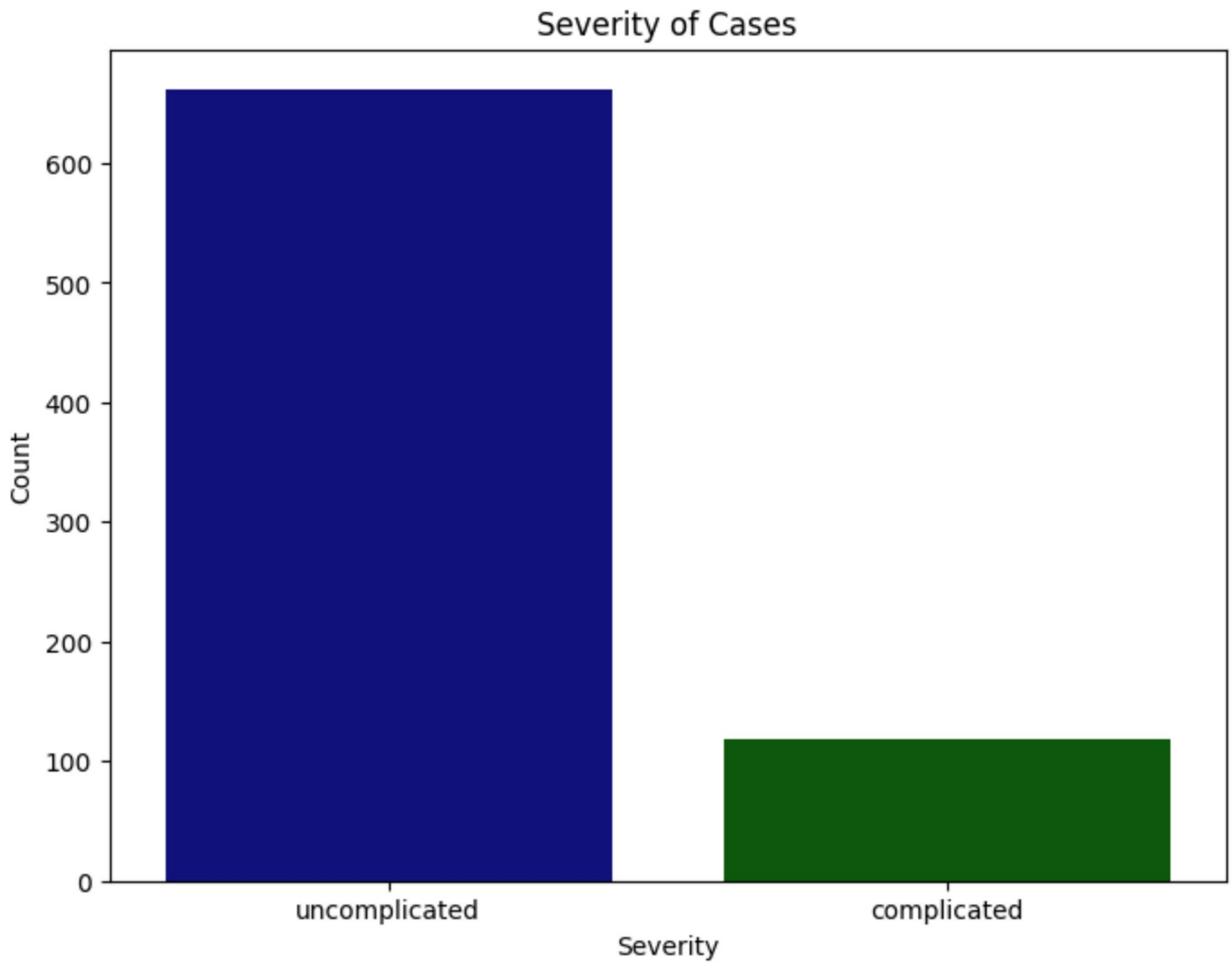


Fig. 6 Patient severity in the Regensburg Pediatric Appendicitis Dataset

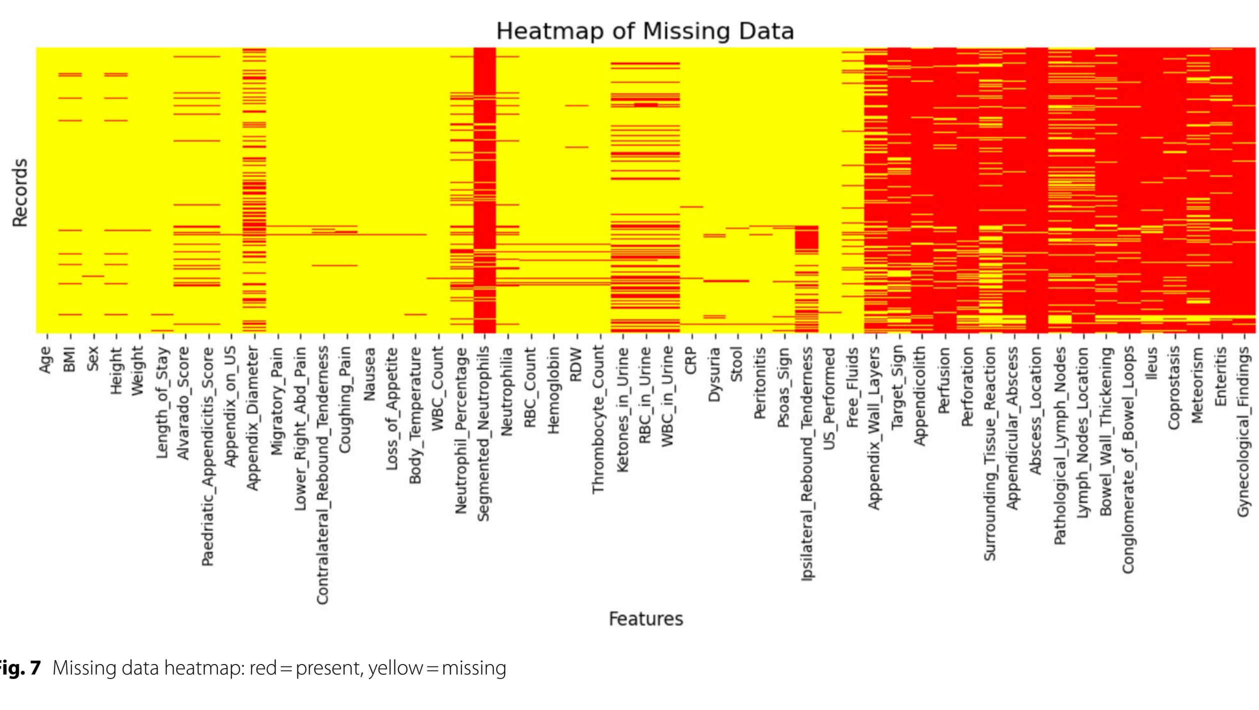


Fig. 7 Missing data heatmap: red = present, yellow = missing

Table 6 Performance comparison of the proposed model, evaluated using F1-score, precision, accuracy and recall

Metric	Score (%)
Accuracy	97.3
Precision	96.5
Sensitivity (Recall)	97.8
Specificity	96.2
F1-Score	97.1
Area Under ROC	96.8

Table 7 Ablation study summarizing the impact of different model components (TabNet, ResNet18, and fusion) on prediction performance

Configuration	Accuracy (%)	AUROC
ResNet18 (Ultrasound only)	84.7	88.1
TabNet (Clinical only)	89.1	91.4
Feature Concatenation (Late Fusion)	89.6	92.4
Full Model with Cross-Attention	**97.3**	**96.8**

moderate contributions, indicating supportive diagnostic value. In contrast, demographic variables and non-specific symptoms exhibited minimal importance, suggesting a limited direct role in model decision-making.

For clarity and visual interpretability, Fig. 8 presents the most influential parameters, while the model was trained using the complete set of 53 clinical features.

Grad-CAM for image feature importance

Gradient-weighted Class Activation Mapping (Grad-CAM) was used to visualize the regions of input images that contributed most to the predictions of the ResNet18 branch (Fig. 9). By backpropagating the gradients from the final convolutional layer, Grad-CAM produces a heatmap that highlights the discriminative image areas.

The Grad-CAM indicates that the CNN model is primarily concentrating on the central and lower abdominal areas of the ultrasound, which are relevant regions for identifying appendicitis. However, the highlighted activation is diffuse rather than sharply defined, suggesting that the model may be depending more on overall image patterns than on precise diagnostic features of the appendix. The comparison between the proposed approach and existing literature is provided in "Comparison with existing approaches" sub-section.

Comparison with existing approaches

The proposed method was benchmarked against existing models from recent literature that used clinical or image-based approaches. As seen in Table 8, performance gains were observed both in overall accuracy and AUROC.

These results confirm the advantage of leveraging both clinical and imaging data with a learnable attention-based fusion strategy. "Statistical significance" sub-section provides the statistical significance of cross-attention fusion in the predictions of the model.

Statistical significance

A McNemar's test was performed to compare the proposed model with the late fusion baseline. The observed difference in misclassification patterns was statistically

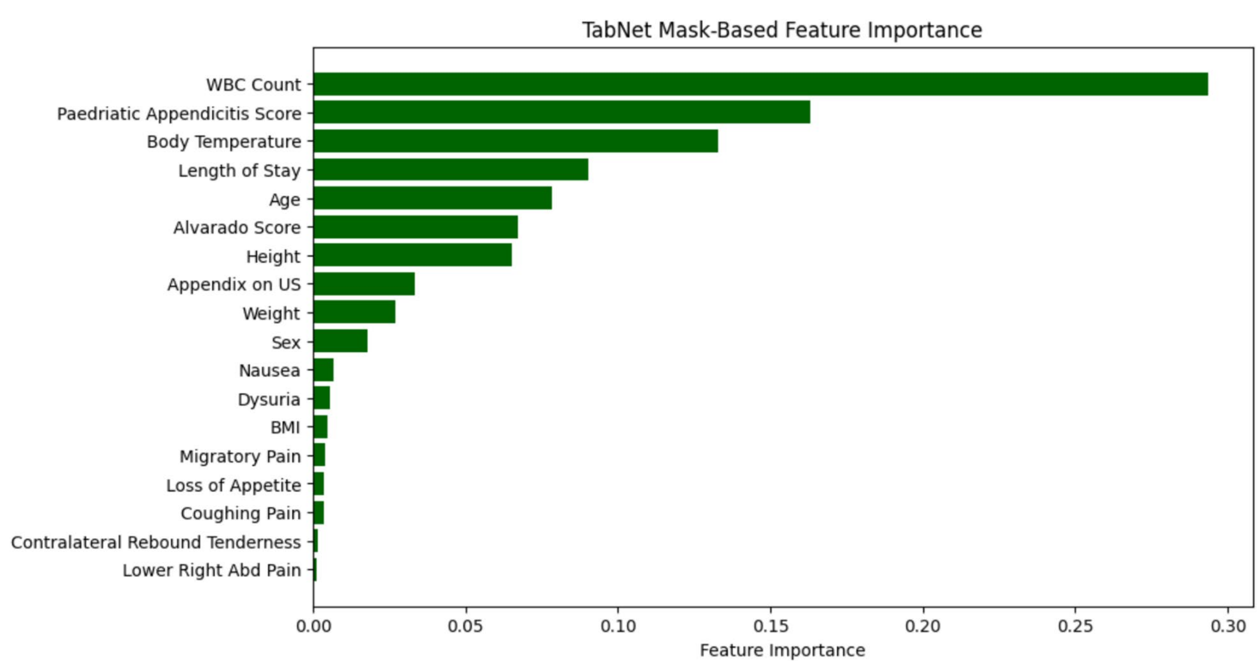
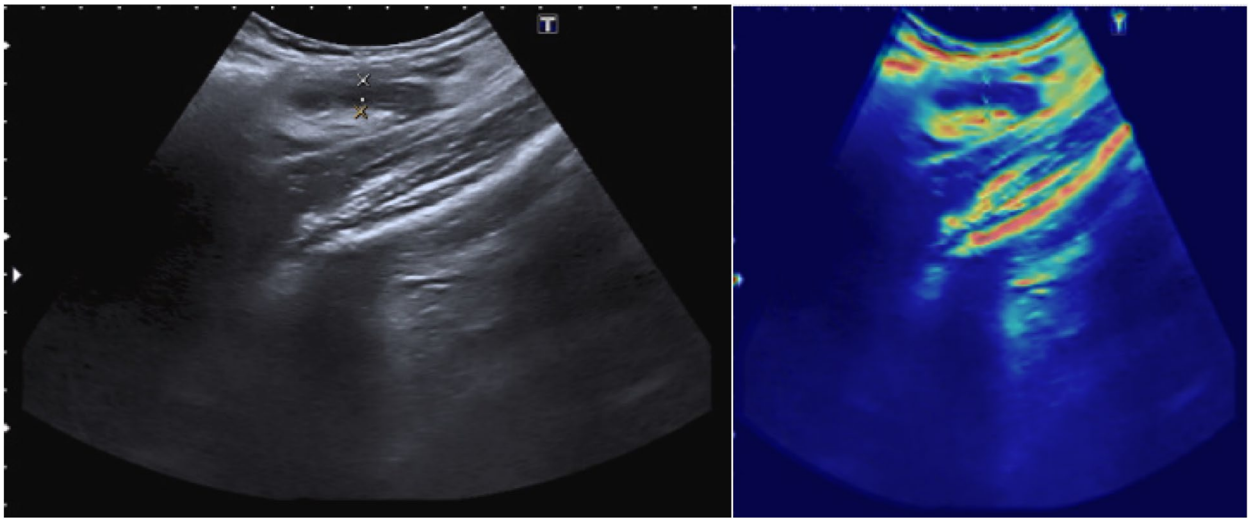


Fig. 8 TabNet mask-based clinical feature importance: WBC count, pediatric appendicitis score and body temperature are the features that contribute most to prediction, followed by length of stay, age and Alvarado score



(a) (b)

Fig. 9 **a** Left: Original ultrasound image used as input to the model. **b** Right: Grad-CAM visualization highlighting the regions most influential in the model's prediction

Table 8 The performance of the proposed model is compared with recent state of the art approaches

S. No	Reference	Study/Model	Key Features	Performance Metrics
1	[13]	Multiple ML Pipelines	Pediatric dataset, clinical + imaging features	Accuracy: 70.1%, AUROC: 0.77, NPV: 82.8%
2	[29]	Concept Bottleneck Model	Ultrasound images + concept supervision	AUROC: 0.80, AUPR: 0.92
3	[32]	XGBoost	Clinical and lab data (severity-based classification)	AUC: 0.914, Accuracy: 85.5%, Sensitivity: 86.5%, Specificity: 84.6%
4	[33]	Logistic Regression, KNN	Age, CRP, PALC; minimal features	Accuracy: 96%, Specificity: 100%, AUC: > 0.90
5	[34]	Stacking Ensemble (with SMOTE–Tomek)	Balanced clinical data, ensemble of multiple classifiers	Accuracy: 92.6%, F1 Score: 92.0%, Recall: 89.1%
6	[35]	CatBoost + SHAP	Interpretable AI; Clinical lab features	AUC ≈ 0.95; Accuracy ≈ 92%; Precision ≈ 93%; Recall ≈ 92%
7	[36]	Alvarado Score	Clinical scoring system based on symptoms, signs, and lab findings	Sensitivity: 89.3%, Specificity: 86.9%, PPV: 71.7%, NPV at 95.6%
8	[37]	LR, RF, SVM, LGBM	Clinical Lab + Imaging features	Accuracy: 94.5%, Precision: 93.8%, Sensitivity: 95.2, AUROC: 0.94
9	[38]	Explainable ML (RF)	Clinical + Imaging features	Sensitivity: 99.55%, Specificity: 14.95%
10	[39]	ML Algorithms (6 models)	pre-operative clinicopathological data	96% accuracy, 60% sensitivity, 100% specificity, NPV 96%
11	[40]	Ensemble/stacking, KNN, DT, bagging, XAI	Clinical, lab, imaging data	Accuracy 92.63%, Precision 95.29%, F1 score = 92.04%
12	[41]	DT	CT imaging, lab, clinical data	AUC 0.70, Sensitivity 85.9%, Specificity 58.5%
13	[42]	XGBoost, SVM, RF, CART	Clinical and laboratory data	AUC = 0.914, Accuracy = 0.855, Sensitivity = 0.865, Specificity = 0.846, NPV = 0.848, PPV = 0.897
14	[43]	Multiple ML algorithms	Clinical, lab data	Accuracy 95.31%
15	Proposed	TabNet, ResNet-18, Cross-attention Fusion	Multimodal fusion with attention	AUC = 0.9, Accuracy = 0.97, Sensitivity = 0.95, Specificity = 0.846

CART Classification and Regression Trees, *CatBoost* Categorical Boosting, *DT* Decision Trees, *GBM* Gradient Boosting Machine, *KNN* K Nearest Neighbour, *LGBM* Light Gradient Boosting Machine, *LR* Logistic Regression, *RF* Random Forests, *SHAP* SHapley Additive exPlanations, *SVM* Support Vector Machines, *XAI* Explainable Artificial Intelligence, *XGBoost* eXtreme Gradient Boosting significant ($p < 0.01$), indicating that the performance improvements are not due to random variation. These findings indicate that fusion enables a richer, more accurate decision based on both visual and structured data. The multimodal approach improves diagnostic accuracy by integrating complementary information, addressing the shortcomings of single-source inputs, strengthening robustness, and better mirroring actual clinical decision-making. A detailed interpretation of the findings and their implications is presented in the following section.

Discussion

The experimental results demonstrate that the proposed multimodal architecture achieves high diagnostic performance in distinguishing complicated and uncomplicated appendicitis. The high sensitivity (97.8%) indicates that the model effectively detects positive cases, which is critical in clinical settings where missed diagnoses can lead to severe complications. The ablation study confirms that both clinical and imaging modalities contribute complementary information. Models trained on a single modality achieved lower accuracy, whereas the cross-attention fusion significantly improved performance. This suggests that the attention mechanism enables more effective interaction between structured clinical features and ultrasound image representations.

The feature importance analysis further indicates that WBC count, pediatric appendicitis score, and body temperature are the most influential predictors. These findings align with established clinical knowledge, where inflammatory markers and clinical scoring systems play a crucial role in appendicitis diagnosis. Similarly, Grad-CAM visualizations show that the CNN branch focuses primarily on the central and lower abdominal regions, which are clinically relevant areas for identifying appendiceal inflammation.

Compared with previous studies summarized in Table 8, the proposed model achieves higher overall accuracy and competitive AUROC values. The improvement can be attributed to the multimodal fusion strategy and cross-attention mechanism, which allow the model to integrate complementary clinical and imaging information. Finally, the statistical significance analysis using McNemar's test ($p < 0.01$) confirms that the observed performance improvement over the late-fusion baseline is statistically meaningful and not due to random variation.

This study is limited by the use of a single multimodal dataset and the absence of external validation, which may restrict generalizability. It focuses on evaluating the feasibility and comparative performance of deep learning models for appendicitis detection under controlled conditions. The subsequent section outlines the study's conclusions and potential future research directions.

Conclusion and future work

In this study, we developed a machine learning based system for appendicitis detection, designed to be both accurate and interpretable. Rather than relying on a single input source, the method combines ultrasound scans with patient clinical data. This multimodal strategy reduces the limitations of single-source analysis and reflects the way physicians evaluate cases in practice. The model demonstrated strong performance across multiple evaluation measures. It reached an accuracy of 97.3%, with precision of 96.5% and sensitivity of 97.8%. Specificity was 96.2%, the F1-score 97.1%, and the AUC 96.8%. Taken together, these results indicate that the model can reliably distinguish between appendicitis and non-appendicitis cases. The high sensitivity points to its usefulness in preventing overlooked diagnoses, while the high specificity suggests that unnecessary procedures can be avoided. An analysis of predictors highlighted WBC count as the most important contributor, followed by pediatric appendicitis score, and body temperature. Overall, the system proves to be a useful decision-support tool that can help clinicians in improving diagnostic confidence for suspected appendicitis. Looking ahead, future work should consider integrating multi-center, multimodal datasets, external validation to improve robustness and generalizability. Real-time implementation techniques may also strengthen the system's clinical value and promote wider adoption.

Authors' contributions

Anurag Barthwal conceived the study, developed the methodology, curated the data, designed the methodology, performed visualization, and led the manuscript writing. Amit Kumar Goel provided guidance throughout the study and assisted in validation. Prasun Chakrabarti offered technical guidance and helped in refining the experimental framework. Sivaneasan Bala Krishnan contributed domain expertise, supported the conceptual development, and reviewed the manuscript to enhance its clarity and rigor.

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Data availability

The dataset used in this study is openly accessible at the UCI Machine Learning Repository and has been cited.

Declarations

Ethics approval and consent to participate

This study was conducted using anonymized, publicly available data, and no ethical approval was required.

Competing interests

The authors declare no competing interests.

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